Machine Learning Project Proposal: Explainable AI

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1 Introduction

Explainable AI (XAI) is increasingly important in today's rapidly advancing technological landscape. Simple models are intrinsically interpretable, like linear models where the magnitude of the weights indicates the importance of the particular feature. But they aren't powerful enough to make predictions. Meanwhile, deep networks are difficult to interpret while being more powerful than linear models. Since we don't want to deny a powerful model just because it's less interpretable, explainable AI helps to make them so while keeping their power. In this project, we are going to interpret the features learned by the model at the hidden layers using different explanation algorithms and evaluate them.

2 Literature Review

XAI have two subtasks, local and global explanation. We will primarily focus on the first one in our project. Local explanation focuses on identifying the most important components that the model relies on to make predictions. Andrea et al.[2] addresses the visualization of image classification model by computing the Saliency Map, where pixels with higher partial derivatives are deemed to be important in prediction. However, this method falls victim to noisy or ill-formatted input, where the hidden components deemed important by the models are not related to the object they try to classify. Daniel et al.[3] introduce SmoothGrad, a gradient-based technique that randomly adds noises to the input image, getting saliency maps of the noisy images, and average them. Due to gradient saturation problem, Mukund et al.[4] advanced the method of Integrated Gradient which ensures sensitivity to all input features that actually affect the output.

3 Objectives

1. For CNN Classifier, we will investigate 5 explanation approaches: Lime Package, Saliency Map [2],Smooth Grad [3], Filter Visualization and Integrated Gradient [4]. If we have time, we will apply modality-specific medical image(MSFI) metric[1] to evaluate how good our gradient-based explanation algorithms are in the medical imaging classification task.
2. For BERT QA pretrained model, we will investigate 3 explanation approaches without evaluation metrics:
   (a) Attention Visualization: Input a sentence to the BERT, and observe the relationship of hidden units across different layers.
   (b) Embedding Visualization: Visualize embedding across layers of BERT using PCA to see whether BERT is trying to cluster similar words together based on the meaning or the context of the words.
   (c) Embedding Analysis: Input different sentences containing the same word, and compare the output embedding of the word across different sentences.

4 Dataset

1. For CNN Classifier's explanation, we use Food 1 dataset, which contains 16643 food images grouped in 11 major food categories. `https://www.kaggle.com/datasets/vermaavi/food11`
2. The BraTS dataset for medical imaging model explanation and evaluation will be from `https://www.med.upenn.edu/cbica/brats2020/data.html` It contains MRI images.
3. The dataset for BERT QA-tuned model's hidden units' explanation will be hand-made sentences.

5 Evaluation Metrics and Expected Outcomes

While many XAI algorithms have been proposed for various tasks, evaluations on their correctness are under-explored. In our project, we will evaluate our explanation algorithms for imaging classification tasks with modality-specific medical image(MSFI) [1] metric, which automates the human assessment process by incorporating the clinical patterns of modality prioritization and feature localization.
We expect the explanation algorithms to output a saliency map for each input image, highlighting the components that are deemed important in final class prediction. We expect the MSFI evaluation metric to give this saliency map a score, the more closely the highlighted area in the saliency map match with the clinical judgment, the higher the MSFI score will be.

References

[1] Weina Jin, Xiaoxiao Li, and Ghassan Hamarneh. Evaluating explainable ai on a multi-modal medical imaging task: Can existing algorithms fulfill clinical requirements? *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(11):11945–11953, June 2022.

[2] Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman. Deep inside convolutional networks: Visualising image classification models and saliency maps, 2014.

[3] Daniel Smilkov, Nikhil Thorat, Been Kim, Fernanda Viégas, and Martin Wattenberg. Smoothgrad: removing noise by adding noise, 2017.

[4] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks, 2017.